

is derived from organic molecules rather than H_2 . Second, instead of occurring in one explosive reaction, respiration uses an electron transport chain to break the fall of electrons to oxygen into several energy-releasing steps (**Figure 9.4b**).

An **electron transport chain** consists of a number of molecules, mostly proteins, built into the inner membrane of the mitochondria of eukaryotic cells (and the plasma membrane of respiring prokaryotes). Electrons removed from glucose are shuttled by NADH to the “top,” higher-energy end of the chain. At the “bottom,” lower-energy end, O_2 captures these electrons along with hydrogen nuclei (H^+), forming water. (Anaerobically respiring prokaryotes have an electron acceptor at the end of the chain that is different from O_2 .)

Electron transfer from NADH to oxygen is an exergonic reaction with a free-energy change of -53 kcal/mol (-222 kJ/mol). Instead of this energy being released and wasted in a single explosive step, electrons cascade down the chain from one carrier molecule to the next in a series of redox reactions, losing a small amount of energy with each step until they finally reach oxygen, the terminal electron acceptor, which has a very great affinity for electrons. Each “downhill” carrier has a greater affinity for electrons than, and is thus capable of accepting electrons from (oxidizing), its “uphill” neighbor, with O_2 at the bottom of the chain. Therefore, the electrons transferred from glucose to NAD^+ , reducing it to NADH, fall down an energy gradient in the electron transport chain to a far more stable location in an electronegative oxygen atom from O_2 . Put another way, O_2

pulls electrons down the chain in an energy-yielding tumble analogous to gravity pulling objects downhill.

In summary, during cellular respiration, most electrons travel the following “downhill” route: glucose $\rightarrow$ NADH $\rightarrow$ electron transport chain $\rightarrow$ oxygen. Later in this chapter, you will learn more about how the cell uses the energy released from this exergonic electron fall to regenerate its supply of ATP. For now, having covered the basic redox mechanisms of cellular respiration, let’s look at the entire process by which energy is harvested from organic fuels.

The Stages of Cellular Respiration: A Preview

The harvesting of energy from glucose by cellular respiration is a cumulative function of three metabolic stages. We list them here along with a color-coding scheme we will use throughout the chapter to help you keep track of the big picture:

1. **GLYCOLYSIS** (color-coded blue throughout the chapter)
2. **PYRUVATE OXIDATION** (light orange) and the **CITRIC ACID CYCLE** (dark orange)
3. **OXIDATIVE PHOSPHORYLATION: Electron transport and chemiosmosis** (purple)

Biochemists usually reserve the term *cellular respiration* for stages 2 and 3 together. In this text, however, we include glycolysis as a part of cellular respiration because most respiring cells deriving energy from glucose use glycolysis to produce the starting material for the citric acid cycle.

As diagrammed in **Figure 9.5**, glycolysis and then pyruvate oxidation and the citric acid cycle are the catabolic pathways that break down glucose and other organic fuels. **Glycolysis**, which occurs in the cytosol, begins the degradation process by breaking glucose into two molecules of a compound called pyruvate. In eukaryotes, pyruvate enters the mitochondrion and is oxidized to a compound called acetyl CoA, which enters the **citric acid cycle**. There, the breakdown of glucose to carbon dioxide is completed. (In prokaryotes, these processes take place in the cytosol.) Thus, the carbon dioxide produced by respiration represents fragments of oxidized organic molecules.

Some of the steps of glycolysis and the citric acid cycle are redox reactions in which dehydrogenases transfer electrons from substrates to NAD^+ or the related electron carrier FAD, forming NADH or FADH_2 . (You’ll learn more about FAD and FADH_2 later.) In the third stage of respiration, the electron transport chain accepts

▼ **Figure 9.4** An introduction to electron transport chains.

